Improving Electrical Schematic Understanding with the ELEC-VQA Dataset: A Comparative Study

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Abstract

Electrical schematic understanding requires interpreting wiring topology under explicit contact-state assumptions. We introduce ELEC-VQA, a dataset derived from computer-aided design drawings of substation secondary circuits. It contains 10,201 English question–answer pairs across 15 task categories and 1,019 images, including 300 original local windows and 719 controlled variants. Each answer is accompanied by concise evidence and graph-based verification. Questions and variants from the same source remain in the same split. Two aligned training sets support a comparison of answer-only and evidence-guided fine-tuning of a seven-billion-parameter vision–language model. We evaluate three general-purpose model families and two fine-tuned models using reference-based semantic assessment. Answer-only fine-tuning improves direct-answer accuracy from 37.16% to 74.27%. With a common explanation prompt, evidence-guided fine-tuning achieves 64.24% joint answer–evidence accuracy, compared with 46.17% for answer-only training; their answer accuracies are 67.97% and 70.16%, respectively. Evidence supervision benefits measurement planning and selected multi-condition questions, but performs poorly on some irrelevant-state changes and topology-changing image pairs. These findings distinguish improvements in explanation quality from improvements in answer accuracy. ELEC-VQA provides a reproducible benchmark for studying both electrical decisions and their supporting evidence. The dataset and code are available at `https://github.com/Talulah81920/ELEC-VQA`.

Keywords: Electrical schematics; evidence-grounded reasoning; visual question answering; domain adaptation.

1 Introduction

Electrical schematics represent connections among terminals, conductors, contacts, and devices. Drawing inspection and maintenance planning require determining whether a contact opening disconnects a branch, which faults explain continuity measurements, or whether a proposed connection satisfies a requirement. These decisions depend on both wiring topology and stated operating conditions. A missed conductor, an incorrectly inferred junction, or an assumed contact state can reverse the answer.

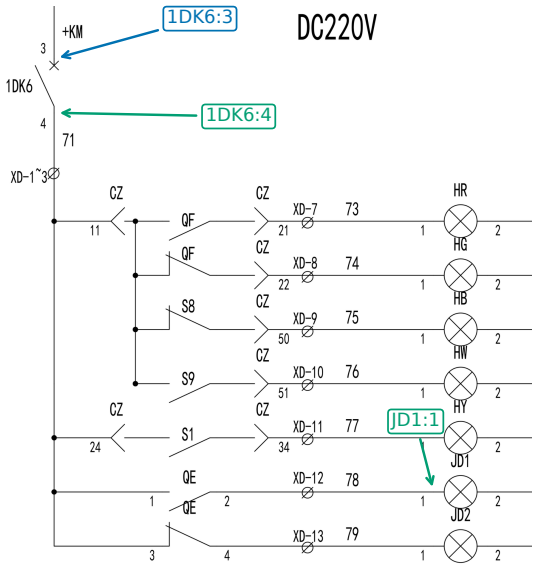
Figure 1 illustrates a measurement-planning example from an original DWG window in ELEC-VQA. The task is to select the fewest continuity measurements that distinguish three single-contact faults. The upstream contact is shared by two branches; the downstream contacts affect separate branches. Measurements at 1DK6:4 and JD1:1 distinguish the three faults, whereas a single binary measurement cannot. The example requires terminal identification, path tracing, and comparison of fault hypotheses, motivating annotations that include both an answer and its supporting evidence.

Visual question answering (VQA) offers a natural-language interface to this information. Document-oriented benchmarks establish the importance of reading text within its visual context [1, 2], while diagram and chart benchmarks examine relations that cannot be recovered from isolated labels [3, 4]. Recent engineering benchmarks extend this direction toward professional artifacts. DrawingVQA evaluates construction drawings through perceptual, contextual, and domain-expert questions [5]; DesignQA examines engineering requirements and documentation [6]; and MechVQA addresses mechanical drawing interpretation [7]. Together, these studies motivate task definitions grounded in engineering practice rather than generic image description.

Electrical diagram understanding is itself an established research area. CircuitVQA, ElectroVizQA, and EEE-Bench cover circuit images, educational electronics, and broader electrical engineering problems, respectively [8–10]. CircuitSense studies visual-to-symbolic circuit analysis [11], and OmniSch evaluates schematic grounding and graph construction [12]. The complementary question considered here is how to construct reusable training and evaluation data for local engineering decisions in native electrical drawings, with inspectable evidence and reproducible state-dependent labels.

Three requirements guide our dataset design. An-

(a) Original DWG circuit window



(b) Engineering question

Using 1DK6:3 as the reference, where should continuity be checked to distinguish the observable fault patterns with the fewest continuity tests? Choose from 1DK6:4, JD1:1, CZ:11, CZ:24, JD2:1 and QE:1. Exactly one of 1DK6 contact 3–4, QE contact 1–2 and QE contact 3–4 is stuck open; the others are closed and wires intact. The circuit is isolated. Treat unlisted contacts as open.

Answer: 2 tests: 1DK6:4 and JD1:1.

(c) Checkable fault signatures

Stuck-open contact	1DK6:4	JD1:1
1DK6 3–4	0	0
QE 1–2	1	0
QE 3–4	1	1

1 = connected to 1DK6:3; 0 = disconnected.

Three distinct signatures; one binary test cannot distinguish three possibilities. Two tests suffice.

Why topology matters: common and branch-specific contacts produce different fault signatures.

Visual grounding → explicit fault conditions → test selection → verifiable evidence

Figure 1: An illustrative ELEC-VQA measurement-planning example from the training partition, using an original DWG window. The released question and answer appear beside compact reference evidence. Continuity tests at 1DK6:4 and JD1:1 give distinct signatures $(0, 0)$, $(1, 0)$, and $(1, 1)$ for the three candidate faults; a single binary measurement cannot distinguish all three. Rendering adjustments and editorial callouts preserve the original wiring topology. The associated graph queries verify the stated fault signatures.

swers must be linked to identifiable electrical entities and checkable paths, blocking conditions, or fault alternatives. Questions must distinguish visible wiring from assumed contact states, because a static schematic does not establish the equipment's current operating state. Evaluation must also account for dependence among questions, overlapping windows, and edited versions of a source circuit.

We introduce ELEC-VQA, derived from native computer-aided design (CAD) drawings. It contains 10,201 English question–answer (QA) pairs and 1,019 images from 44 DWG files across 14 substation projects. The 300 original windows and 719 controlled variants support conditional connectivity, fault diagnosis, measurement planning, and local design checks. Each answer includes concise evidence linked to terminal identifiers, source geometry, and executable graph queries. Reference graphs support annotation generation and verification; the VQA models receive only images and questions.

Using the same seven-billion-parameter vision–language backbone, we compare fine-tuning on short answers with fine-tuning on evidence followed by the same answers. Both use low-rank adaptation (LoRA) [13]. Evidence consists of explicit intermediate statements that can be checked against the circuit representation. We examine where this supervision improves or reduces performance, without treating generated explanations as observations of internal reasoning.

The study makes three contributions to domain-specific schematic understanding:

1. A CAD-derived electrical schematic dataset with 15 task families, concise English questions and answers, compact evidence, and terminal-level provenance.
2. A construction and verification protocol combining native rendering, controlled wiring interventions, executable checks, and source-disjoint partitions.
3. A comparative evaluation of general-purpose models and aligned fine-tuned models, separating answer accuracy from evidence quality.

The empirical scope is substation secondary wiring. Section 2 reviews related work, Section 3 describes

dataset construction, and Sections 4 and 5 present the methods and results.

2 Related Work

2.1 Document and Diagram Question Answering

TextVQA introduces questions requiring recognition and contextual use of text in images [1]. DocVQA transfers question answering to document pages, where layout and structure determine how textual evidence should be interpreted [2]. ChartQA and InfographicVQA extend the problem to joint interpretation of text, graphics, and quantitative relations [4, 14]. These resources establish that reading and reasoning should be assessed together, but their document semantics differ from the conductor and contact relations of an electrical drawing.

AI2D represents diagram elements and their relations for diagram understanding [3]. GQA uses structured scene representations to generate compositional questions and evaluate consistency [15]. These approaches motivate explicit representations and relational checks, although a scene graph does not by itself express electrical continuity under changing contact states. MMMU provides a broader test of multimodal subject knowledge, including engineering [16]; a specialized benchmark is needed to resolve failures at individual terminals and connections.

2.2 Engineering Drawings and Professional Questions

DrawingVQA evaluates construction-drawing understanding using 33 drawings and 92 curated QA pairs at perceptual, contextual, and domain-expert reasoning levels [5]. It motivates questions grounded in engineering practice. ELEC-VQA instead focuses on electrical connectivity under explicit contact states, with CAD-derived graphs for annotation verification. The different domains and annotation procedures preclude direct cross-dataset accuracy comparisons.

DesignQA connects engineering documentation, visual artifacts, and requirement checking [6]. Its emphasis on applying requirements is relevant to our bounded design-check tasks, where requirements are stated in the question rather than inferred from an unstated standard. Work in IEEE Access on multimodal document segmentation further links technical QA to the organization of local visual elements [17]; our benchmark supplies a fixed local image rather than evaluating document retrieval. MechVQA combines mechanical drawings with recognition, reasoning, and judging tasks and domain adaptation [7]. These studies support a dataset-centered comparative design: the contribution lies in the representation of domain decisions, annotation provenance, and diagnostic evaluation, not merely in the number of image-question pairs.

2.3 Electrical Schematics and Circuit Reasoning

Earlier graph-based circuit-sketch recognition addresses the structured interpretation of drawn circuit elements [18]. CircuitVQA addresses components, values, junctions, and positions in circuit images [8]. ElectroVizQA focuses on digital electronics problems [9], whereas EEE-Bench spans multiple electrical and electronic engineering subdomains [10]. CircuitSense emphasizes deriving symbolic circuit models from visual inputs [11]; OmniSch focuses on grounding schematic entities and recovering spatially weighted netlist graphs [12]. Work on near-sight correction further considers local visual inspection in circuit understanding [19]. These directions show why a response-level score alone cannot localize whether an error arose from reading, topology extraction, or subsequent reasoning.

ELEC-VQA complements these resources with native industrial wiring images, explicit contact assumptions, and brief verifiable explanations. Its tasks concern visible external connections, not unrestricted analog simulation, complete device behavior, or undocumented internal wiring. Table 1 summarizes the scope and verification emphasis of these resources; it is a descriptive comparison rather than a cross-dataset performance ranking.

2.4 Evidence Supervision and Evaluation

ScienceQA includes explanatory supervision for multimodal questions [20], and A-OKVQA provides knowledge-based visual questions with rationales [21]. Multimodal-CoT separates rationale generation from answer inference [22]; Distilling Step-by-Step trains smaller language models using teacher-generated rationales [23]. These studies motivate intermediate supervision. In ELEC-VQA, however, evidence is derived from CAD graphs and rules rather than a separately trained teacher model.

Explanation evaluation requires a separate notion of support. Multimodal explanation research distinguishes textual justification from visual evidence [24], and ERASER distinguishes rationale alignment from faithfulness [25]. DRAGON evaluates evidence localization over diagrams with the correct answer supplied [26]; our primary task instead requires predicting the answer from an image and question. Generated reasoning may also omit factors that actually influenced a prediction [27]. We therefore evaluate explicit evidence as a verifiable explanation, without treating fluent prose as proof of internal reasoning.

Visual instruction tuning and modern open vision–language backbones provide the implementation basis [28–30]. Position, verbosity, and self-preference effects documented for language-model judges [31] motivate blinded scoring, structured references, and independent reliability checks.

LAVE evaluates generated VQA answers through

Table 1: Related resources and their reported emphasis. Descriptions concern their task definitions, not results from a common evaluation.

Resource	Source artifact	Question emphasis	Evidence / verification	Evaluation unit
DrawingVQA [5]	Construction drawings	Perception to expert judgment	Expert-curated QA / reasoning-depth labels	Drawing / question
DesignQA [6]	Engineering documents and visual artifacts	Requirements and compliance	Document-grounded references	Requirement / question
MechVQA [7]	Mechanical drawings	Recognition, reasoning, judging	Semi-automated quality-control pipeline	Drawing / question
CircuitVQA [8]	Circuit images	Components, values, junctions, positions	Circuit-specific QA annotations	Image / question
EEE-Bench [10]	Electrical and electronic problems	Broad engineering problem solving	Curated problem references	Problem
CircuitSense [11]	Schematics and block diagrams	Perception, analysis, design	Symbolic-equation references	Problem / hierarchy
OmniSch [12]	Printed circuit board (PCB) schematics	Grounding and graph construction	Entity / netlist graph annotations	Diagram / structured task
ELEC-VQA (Ours)	Native substation DWGs	State-dependent local wiring decisions	CAD graph checks + concise evidence	QA / image; source-grouped split

reference-based language-model assessment [32]. Prior work also evaluates Qwen2.5-32B-Instruct as a judge [33]. We adapt this approach with electrical criteria covering terminal identities, contact states, complete solutions, and valid alternatives. Prior judge benchmarks do not establish reliability on electrical schematics.

BERTScore compares contextual token representations [34], whereas electrical correctness additionally depends on terminal identities and state assumptions. G-Eval motivates structured evaluation criteria [35]; LLMBar demonstrates that persuasive responses can mislead evaluators [36]. We therefore use task-specific criteria rather than textual similarity alone.

3 ELEC-VQA Dataset Construction

3.1 Sources, Scope, and Sampling Units

The ELEC-VQA dataset contains 10,201 QA records from 44 DWGs in 14 substation projects. Its 1,019 image instances comprise 300 original local context windows and 719 controlled native-DWG variants. Original windows include surrounding schematic context and can overlap within a source drawing. Each record retains project, drawing, parent-window, and image-configuration identities, allowing dependent observations to be grouped during partitioning and analysis.

The intended unit of model input is a readable local schematic, rather than a complete drawing sheet. Selection favors windows whose visible external wiring and explicit contact pairs admit a bounded representation. Questions are not restricted to label transcription, but symbol and label recognition remain prerequisites for higher-level tasks. Figure 2 shows the construction overview, and Table 2 reports the partition summary.

3.2 Native Rendering and Source Traceability

Source geometry is extracted through the AutoCAD ActiveX Component Object Model (COM) interface. Font-corrected working copies are rendered to vector PDF and rasterized using recorded CAD-to-image mappings. A 0.35-mm lineweight improves conductor visibility without changing coordinates or junction topology. No raster dilation or generative redrawing is used.

Terminal positions, conductor geometry, source identifiers, crop bounds, and image hashes support traceability. Controlled faults are applied in native working copies and rendered under registered views. Figure 3 shows registered crops from an original window and a controlled cross-network link, with their reference graphs. Display crops are distinguished from the full released images supplied to models.

3.3 State-Dependent Electrical Representation

A local circuit is represented as $G = (V, E_w, E_c)$, where V contains terminals and junctions, E_w contains external conductors, and E_c contains explicitly modeled contact pairs. For a state assignment s, removed-wire set R, and optional jumper set J, the active graph is

$$G_{s,R,J} = (V, (E_w \setminus R) \cup \{e \in E_c : s(e) = 1\} \cup J)\,. \tag{1}$$

Reachability in this graph determines connectivity within the local circuit. A contact's drawn appearance is not substituted for the state stipulated in the question. For a relevant unknown state, admissible assignments are evaluated separately; the answer may

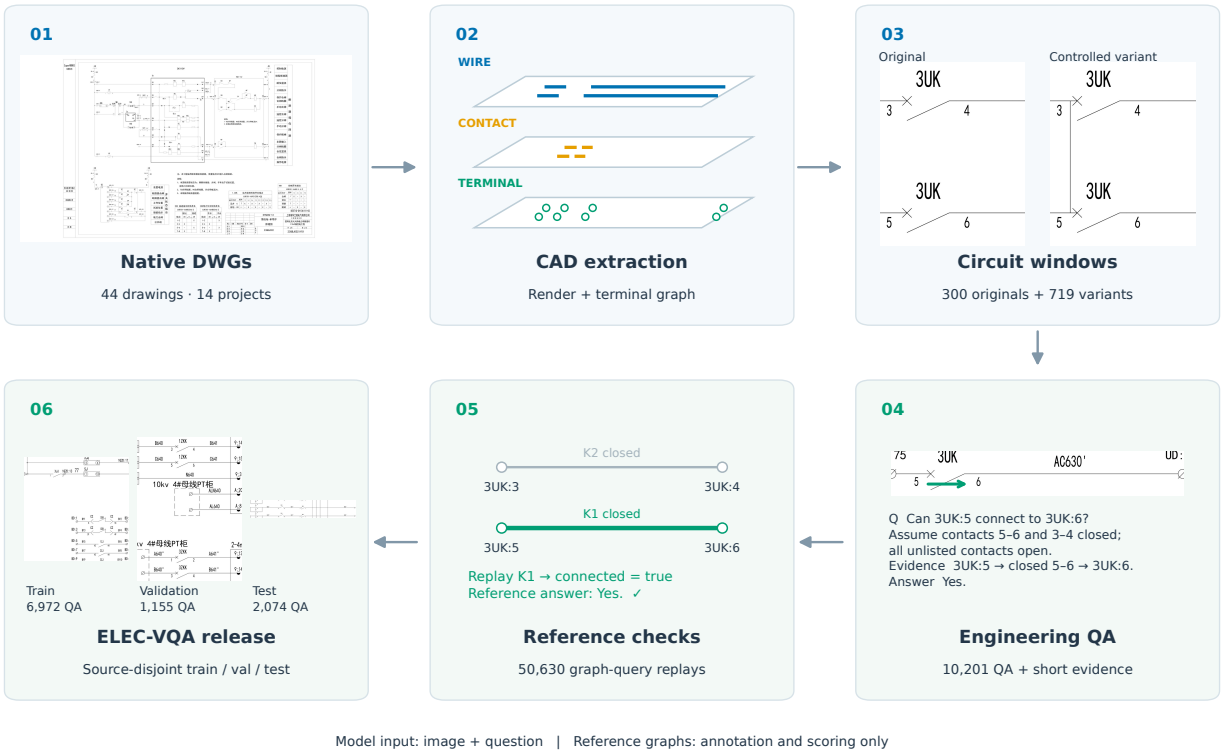


Figure 2: Source-traceable dataset construction. The model receives an image and a question; source graphs and verification queries remain annotation-side information.

Table 2: Source-disjoint partitions. Windows count original contexts; variants count controlled native edits.

Split	Proj.	DWGs	Windows	Variants	Images	QA
Train	11	29	218	479	697	6,972
Validation	1	8	34	88	122	1,155
Test	2	7	48	152	200	2,074
Total	14	44	300	719	1,019	10,201

depend on that state. A graphical crossing is not automatically a junction, and undocumented device internals are excluded.

The representation does not establish actual voltage, current magnitude, protection operation, or system safety. Diagnosis assumes the stated fault family, and continuity measurements refer to isolated circuits with intact wires unless otherwise specified. These restrictions make the answer checkable; they also delimit the engineering claim.

3.4 Questions, Answers, and Compact Evidence

Fifteen task families cover conditional connectivity (T01), minimum closure (T02), wire groups (T03), contact-opening effects (T04), unknown states (T05), bypasses (T06), fault candidates (T07), measurement selection (T08), stated design requirements (T09), repair checks (T10), path order (T11), contact-network relationships (T12), diagnostic sufficiency (T13), voltage-information sufficiency (T14), and branch interruption (T15). Table 3 reports their distribution. The categories describe task content rather than difficulty levels.

Questions use circuit-aware templates and controlled wording variants. Their language is English except where a source annotation must be quoted. A single image supports different decisions rather than repeated paraphrases counted as new semantic cases. Each master record separates the short answer, typed result, compact evidence steps, source entities, and verification queries. The evidence states the relevant condition, a conducting path or blocking fact, and only the alternatives necessary to justify the conclusion. Paths provide evidence of connectivity; negative statements retain the solver's reachable region in the structured record rather than claiming that an omitted route proves disconnection.

Evidence text is limited to 192 tokens under the local Qwen tokenizer, excluding headings and the answer; the published maximum is 189 and the mean is 39.63. Over-budget candidates are excluded rather than truncated. Figure 4 shows the task, evidence-

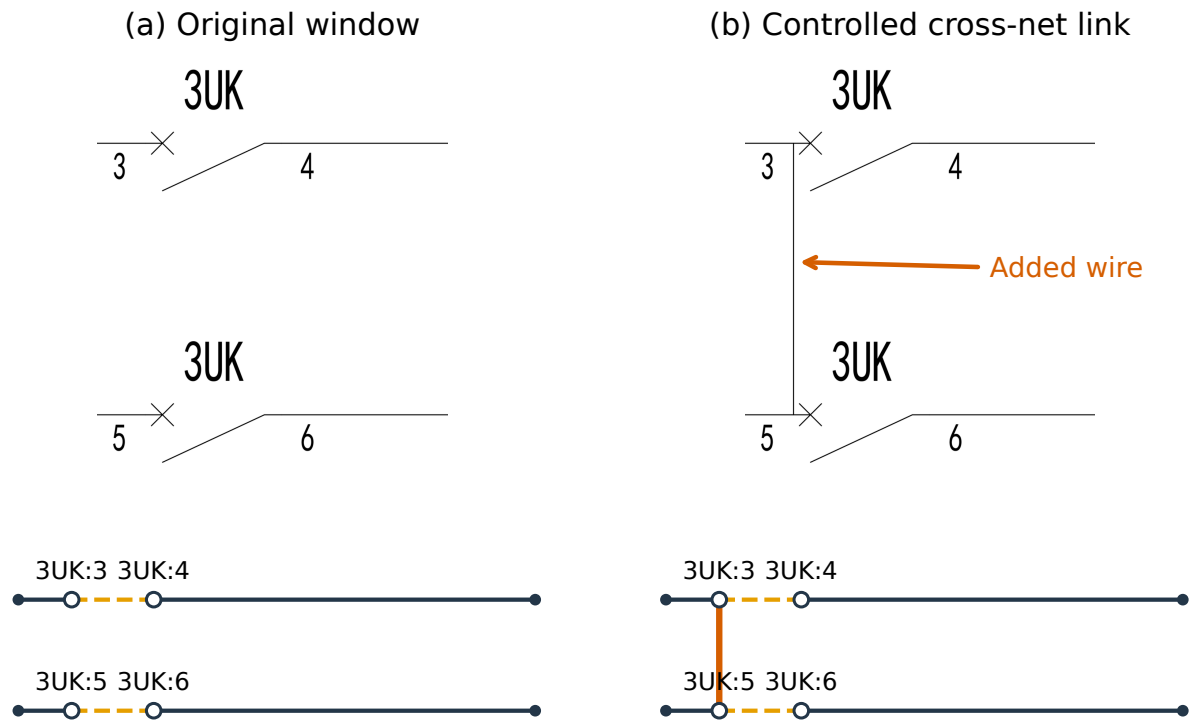


Figure 3: Original local window and controlled native-DWG cross-network link. Registered crops show the added conductor connecting 3UK:3 and 3UK:5; the bottom panels show the corresponding bounded reference graphs. Dashed graph edges denote modeled contacts whose states are given by the question. Cropping and the arrow are for display only; models use the full released images.

length, and image-provenance distributions. T12 currently contains series and separate-network cases but no verified parallel examples. T14 contains only negative sufficiency answers and is treated as a diagnostic control, not evidence of learned visual reasoning.

3.5 Interventions, Verification, and Partitions

The 719 controlled variants include omitted wires, contact bypasses, incorrect connections, and cross-network links. They are synthetic edits, not observed field-fault frequencies. Tasks are instantiated under explicit contact-state assignments and fault assumptions, with candidate questions retained only when their reference outcomes and evidence satisfy the corresponding verification queries.

The dataset audit replays 50,630 graph queries and checks training-set alignment, duplicate task specifications, file hashes, and split membership. Duplicate checks compare task parameters and verification conditions within each image and task; they do not establish graph-isomorphism independence. Image decoding and geometry checks verify rendering consistency. Shared extraction errors may nevertheless produce self-consistent reference annotations.

Training, validation, and test contain 6,972, 1,155, and 2,074 QA over 697, 122, and 200 images, respectively. All source projects, DWGs, and parent families are disjoint across splits. Overlapping context windows and related variants remain grouped, but electrically similar templates may occur in different projects. Reporting follows the principle that composition, provenance, and reuse conditions should be documented explicitly [37]. The evaluation scope and reference-dependent limitations are specified in Section 5.

4 Evidence-Guided Adaptation and Evaluation

4.1 Aligned Supervision Views

For image I_i, question q_i, short answer a_i, and compact evidence e_i, the two principal targets are

$$y_i^A = a_i, \qquad y_i^E = \texttt{Evidence:}\ \ e_i\ \texttt{Answer:}\ \ a_i. \quad (2)$$

Both training sets use identical sample IDs, questions, images, and final answers. Reference graphs and verification metadata are excluded from model inputs. Figure 5 illustrates a diagnostic example: either of two series-contact faults explains the failed continuity measurement. The evidence identifies both alternatives, distinguishing fault compatibility from unique diagno-

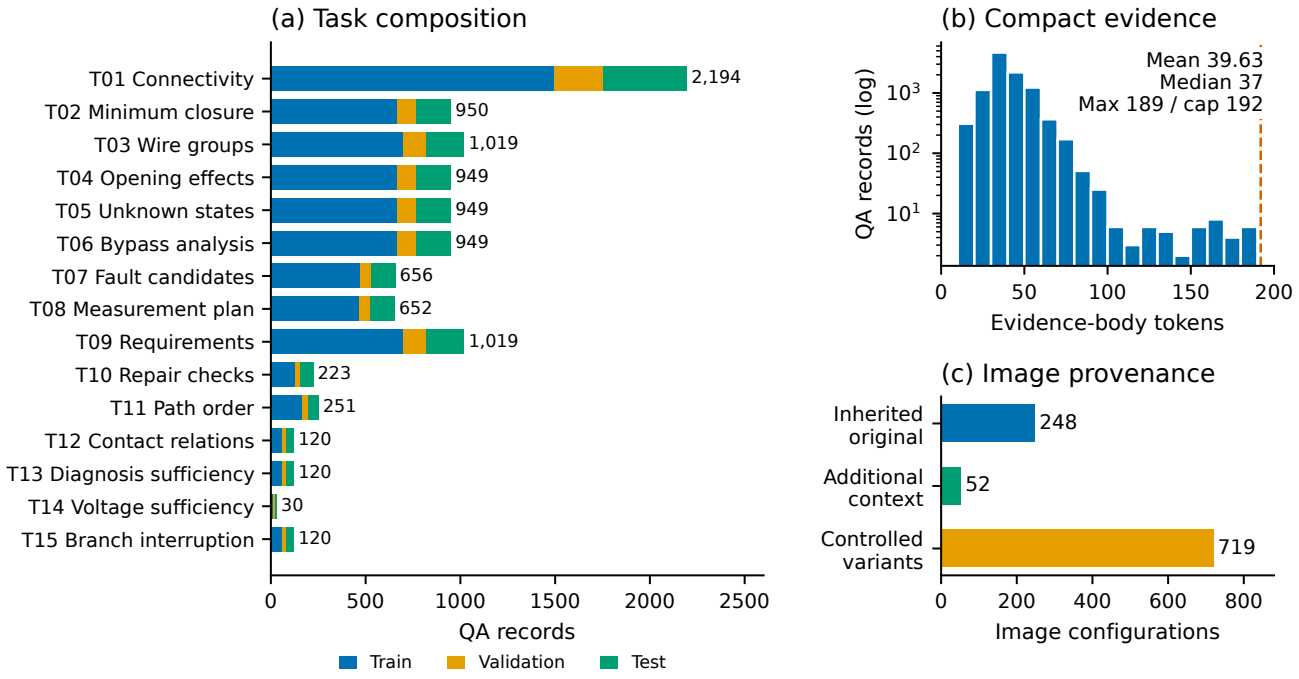


Figure 4: Composition of the ELEC-VQA dataset: (a) QA by task and split; (b) evidence-body length under the local Qwen tokenizer, with logarithmic count scale and a 192-token cap; and (c) image provenance. Original windows may overlap within a drawing; controlled variants retain their parent-window identities.

Table 3: QA counts by task and split. T14 is a diagnostic control, and T12 contains no verified parallel cases.

ID	Task family	Train	Val.	Test	Total
T01	Connectivity	1,495	259	440	2,194
T02	Minimum closure	666	103	181	950
T03	Wire groups	697	122	200	1,019
T04	Opening effects	665	103	181	949
T05	Unknown states	665	103	181	949
T06	Bypass analysis	665	103	181	949
T07	Fault candidates	471	58	127	656
T08	Measurement plan	467	58	127	652
T09	Requirements	697	122	200	1,019
T10	Repair checks	130	24	69	223
T11	Path order	164	30	57	251
T12	Contact relations	60	20	40	120
T13	Diagnosis sufficiency	60	20	40	120
T14	Voltage sufficiency	10	10	10	30
T15	Branch interruption	60	20	40	120
Total		6,972	1,155	2,074	10,201

sis. The comparison therefore tests the supervision format under matched inputs.

The backbone is Qwen2.5-VL-7B-Instruct [29]. Standard LoRA [13] updates language attention query, key, value, and output projections; pretrained backbone and vision weights remain frozen. The training loss is computed over assistant-response tokens:

$$\mathcal{L} = -\sum_{i} \sum_{t \in \mathcal{A}_i} \log p_\theta(y_{i,t} \mid I_i, q_i, y_{i,<t}), \qquad (3)$$

where $\mathcal{A}_i$ excludes user, image, and padding positions. All assistant target tokens receive equal loss weight. Because the evidence target is longer, equal epochs do not imply equal supervised-token exposure or computation.

Fine-tuning uses bfloat16 (BF16), rank 16, scaling 32, dropout 0.05, learning rate 10^{-4}, three epochs, microbatch one, and gradient accumulation eight. Image area and context length are capped at 2,097,152 pixels and 8,192 tokens. The slow Qwen processor and scaled dot-product attention (SDPA) are fixed. Runs use Ubuntu 22.04 and one NVIDIA H100 graphics processing unit (GPU). The primary comparison uses seed 42 and the final-epoch adapters. Table 4 summarizes the training settings.

4.2 Reference-Based Semantic Assessment

Exact match can underestimate correct free-form answers and overlook small but decisive changes in terminal labels or contact states. Following LAVE [32], we use reference-based semantic evaluation. The judge receives the question, complete response, reference answer, task rubric, and structured reference information, including relevant graph entities and verification queries. These annotations are used only for evaluation; the judge does not independently interpret the schematic image.

Ratings concern whether the final answer satisfies the electrical task rather than resemblance to the reference wording. The task rubric preserves terminal identities, negation, explicit conditions, set completeness, and path order; alternative valid minimum plans or conducting paths are admissible. The same request

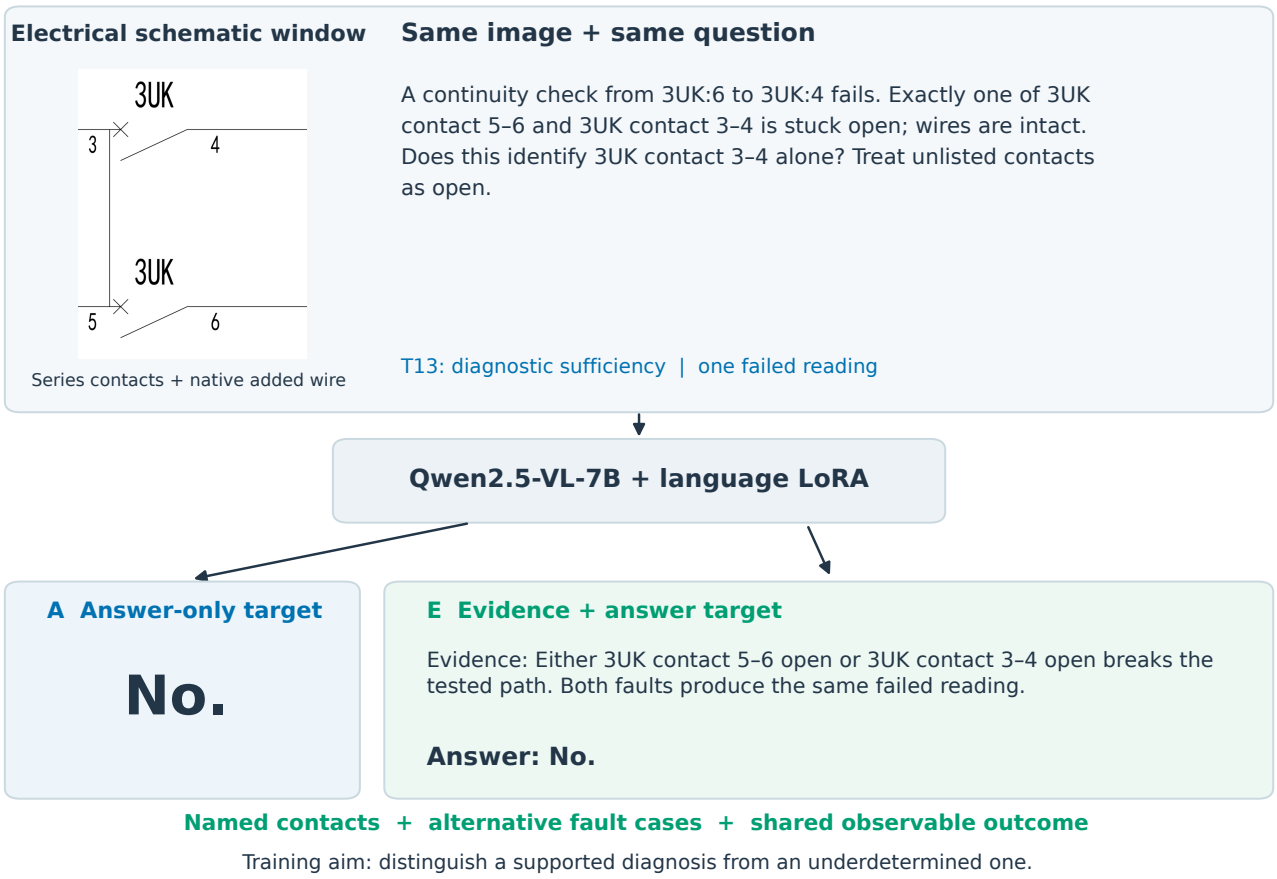


Figure 5: Aligned supervision illustrated by a diagnostic-sufficiency example from the ELEC-VQA dataset. The same image and exact released question lead to either the short answer (A) or the released compact evidence followed by the same answer (E). The evidence identifies two distinct contact faults that produce the same failed continuity reading. Targets are reference annotations, not model predictions; the screenshot is cropped only for display. Equal examples and epochs do not imply equal token exposure or computation.

construction and rubric apply to Base, A, E, and the cross-family baselines.

Let $r_i \in \{1, 2, 3\}$ denote the final reference-based rating. A rating of 3 denotes a fully correct answer, 2 a correct but incomplete answer when permitted by the task rubric, and 1 an incorrect or absent answer. The auxiliary semantic score and full-credit indicator are

$$s_i = (r_i - 1)/2, \qquad c_i = \mathbb{1}[r_i = 3]. \tag{4}$$

Only full credit contributes to the primary accuracy. Binary decisions, three-way state classification, and single relation categories do not receive partial credit. All test responses receive final ratings; no response is excluded because of an evaluation-format failure. The final scoring protocol and a sensitivity analysis using the initial Qwen outputs are described in Section 5 and the supplementary material.

Answer correctness and evidence quality are evaluated separately. A correct answer does not require an explanation. Incorrect supporting details affect evidence and joint scores, unless they also contradict the final answer. Primary results report sample-level accuracy and equally weighted macro averages across 14 task categories. The ten T14 test controls are reported separately, leaving 2,064 questions in the primary test set. Reference quality and evaluator bias remain limitations [31, 36].

4.3 Controlled-Change and Evidence Metrics

Visual pairs use identical questions on different images, whereas condition pairs use the same image with a changed contact-state assumption. The pair audit checks question identity and reference outcomes, retaining both outcome-flip and invariant pairs.

For an admissible pair set $\mathcal{P}$, both responses must receive full credit; let $S_i = c_i$:

$$A_{\text{pair}} = \frac{1}{|\mathcal{P}|} \sum_{(i,j)\in\mathcal{P}} S_i(\hat{a}_i) S_j(\hat{a}_j). \tag{5}$$

A changed answer alone is insufficient: both responses must be correct under their respective images and conditions. We distinguish pairs whose reference answer changes from those whose answer remains invariant. Shared questions and circuit sources are retained in the analysis.

Table 4: Matched training configuration for answer-only and compact-evidence fine-tuning.

Setting	Answer only	Evidence–answer
Backbone	Qwen2.5-VL-7B-Instruct	Qwen2.5-VL-7B-Instruct
Training QA / images	6,972 / 697	6,972 / 697
Target format	Short answer	Compact evidence, then answer
Adapted projections	Language q, k, v, o	Language q, k, v, o
LoRA rank / scaling / dropout	16 / 32 / 0.05	16 / 32 / 0.05
Precision / attention	BF16 / SDPA	BF16 / SDPA
Learning rate / schedule	10^{-4} / cosine	10^{-4} / cosine
Epochs	3	3
Microbatch / accumulation	1 / 8	1 / 8
Maximum pixels / context tokens	2,097,152 / 8,192	2,097,152 / 8,192
Processor / primary seed	Slow / 42	Slow / 42
Checkpoint policy	Final epoch	Final epoch

Evidence is assessed for entity grounding, route or blocking-fact validity, compliance with stated conditions, treatment of necessary alternatives, and support for the answer. Joint correctness requires both a correct answer and valid evidence. Alternative valid explanations are accepted rather than matched verbatim. These criteria concern the observable explanation, not faithfulness to hidden computation [25, 27].

4.4 Common Inputs and Inference Budgets

The controlled comparison contains the unchanged backbone (Base), the answer-only adapter (A), and the evidence–answer adapter (E). InternVL2_5-8B and LLaVA-OneVision Qwen2 7B provide cross-family zero-shot context [30, 38, 39]. Every comparator receives the same image and question under its native processor. Pixel budgets, tiling, visual tokens, checkpoint revisions, and prompts are logged; native processing is not described as pixel-identical preprocessing.

The direct-answer prompt appends "Give a brief answer in English. Do not include an explanation." to each released question. For Base/A/E, a second prompt requests relevant components, connections, and conditions, followed by an `Answer:` line. Complete responses are scored without requiring a particular output format. Both settings use greedy decoding and a 768-token output limit fixed on validation before testing.

4.5 Use of AI-Assisted Tools

OpenAI ChatGPT/Codex assistants [40] supported dataset construction, experimental code development, manuscript preparation, and English-language editing. Supplementary reference-based assessment is documented in Section 5 and the supplementary material. The authors are responsible for the final annotations, analyses, and manuscript. AI assistance is not treated as independent expert validation.

5 Experiments and Comparative Study

RQ1. Cross-model understanding. How do general-purpose vision–language models perform on ELEC-VQA?

RQ2. Domain adaptation. How do answer-only and evidence–answer supervision interact with output prompting?

RQ3. Verifiable evidence. When does evidence supervision improve supported answers?

5.1 Experimental Protocol and Metrics

The dataset contains 6,972 training, 1,155 validation and 2,074 test QA records. Excluding ten T14 diagnostic controls per evaluation split gives 1,145 validation and 2,064 test questions. The test panel spans seven source DWGs from two projects. Questions and variants sharing a source remain statistically dependent. The comparators are Qwen2.5-VL-7B-Instruct (Base), its answer-only adapter A, its compact-evidence–answer adapter E, InternVL2_5-8B and LLaVA-OneVision Qwen2 7B. Adapters use seed 42. Models receive identical questions and raw images with native preprocessing. Greedy decoding allows 768 new tokens. The direct track requests a brief answer without explanation. A second track requests visible components, connections and conditions, followed by a brief final answer, identically for Base/A/E.

Qwen2.5-32B-Instruct is the primary judge, following the reference-based evaluation approach of LAVE [32] and prior Qwen judge studies [33]. Model identities are withheld. Answers and evidence are scored separately using the criteria in Section 4; joint accuracy requires both to receive rating 3 for the same response. Judge inference uses BF16/SDPA, a 12,288-token input limit, 512 output tokens, and one format-recovery retry.

Final scores retain all non-null Qwen ratings and include 585 supplementary judgments: 28 direct-answer, 75 explanation-answer, and 482 evidence ratings, covering 552 responses to 517 questions. All main anal-

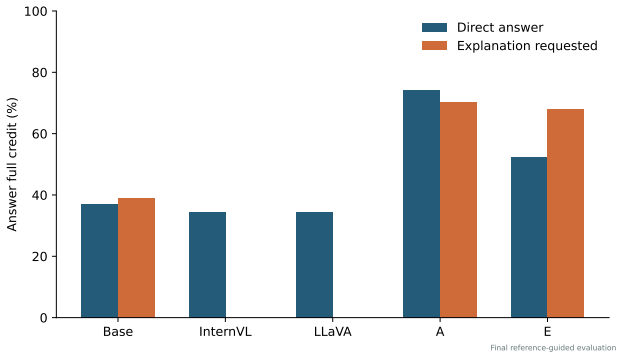


Figure 6: Answer full credit under the two output protocols. Explanation tracks are evaluated for Base/A/E only.

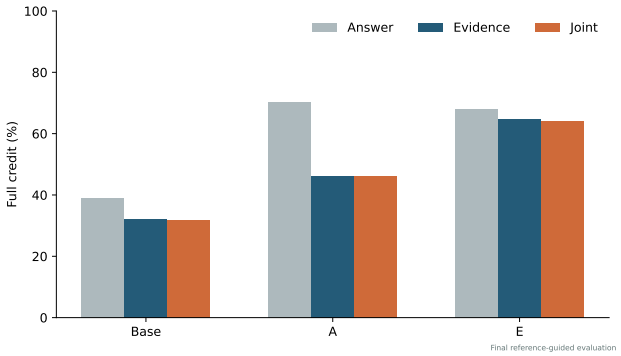


Figure 7: Answer, evidence and joint accuracy with common explanation prompting.

yses use this completed score set. The supplementary material documents provenance and sensitivity to the additional judgments. Both stages remain subject to reference errors and evaluator bias [31]. Accuracy is the proportion of responses rated fully correct. We also report equally weighted task and drawing macro averages and mean semantic scores.

5.2 Cross-Model Comparison

A achieves 74.27% direct-answer accuracy, exceeding Base by 37.11 percentage points. E reaches 52.42%, above Base but below A by 21.85 points. Cross-family baselines remain near 34.5% (Table 5). Native preprocessing and pretraining differences prevent interpreting the comparison as an isolated architectural effect.

5.3 Output Prompt and Evidence Supervision

Under common explanation prompting, answer accuracy is 39.00% for Base, 70.16% for A and 67.97% for E (Fig. 6). E improves by 15.55 points relative to direct mode, with a drawing-cluster bootstrap interval of [12.63,17.86]. A changes by -4.12 points [-4.92,-2.35]. The A-minus-E answer-accuracy gap under explanation prompting is 2.18 points [-1.61,6.65]. An interval spanning zero establishes neither equivalence nor a stable answer advantage for E. Output alignment and additional generated computation remain alternative explanations; these comparisons do not demonstrate a causal effect of visible reasoning text.

5.4 Evidence Quality and Joint Accuracy

E achieves 64.78% evidence accuracy and 64.24% joint accuracy, compared with A's 46.27% and 46.17%. The joint improvement is 18.07 points, with a drawing-cluster bootstrap interval [16.04,20.44]. This supports improved reference-supported responses rather than uniformly better final decisions.

The initial-Qwen sensitivity analysis preserves the main ordering. Direct-answer lower estimates are 74.18% for A and 52.37% for E. For joint accuracy, E's lower estimate (63.76%) exceeds A's upper estimate (63.18%) when supplementary judgments are withheld. These assignment bounds do not account for systematic reference errors; details are provided in the supplementary material.

5.5 Conditions Under Which E Outperforms A

T14 is excluded from Table 7 and the primary aggregates because it is a voltage-information sufficiency control with ten all-negative test answers. A constant "No" predictor could obtain full accuracy without interpreting the image. T14 is therefore reported separately, leaving 14 task categories and 2,064 questions in the primary test set.

E improves answer accuracy on measurement planning (T08, +42.52 points), diagnostic sufficiency (T13, +15.00), open-contact impact (T04, +6.63), branch interruption (T15, +7.50) and contact order (T11, +5.26). Conversely, minimum closure, wire grouping and hypothetical bypass remain weaker. Table 7 reports every primary family. Differences are exploratory and are not adjusted for multiple comparisons. An original-DWG case asks whether bridging open 2ZJ contact 24–23 connects 3UK:4 to CZ:41 while 3UK contact 3–4 is closed. Both adapters fail under direct prompting; A also fails when explanation is requested. E identifies the visible path 3UK:4 → 3UK:3 → 2ZJ:24 → 2ZJ:23 → CZ:41 and answers correctly. The case illustrates a correct explanation derived from a conducting path; it does not establish causal faithfulness.

5.6 Conditional Evidence Validity

Following the distinction between task performance and explanation quality in e-ViL [41], let C_i and V_i indicate whether the answer and evidence receive rating 3, respectively. Evidence validity conditional on a correct answer is

$$Q_{E|A} = \frac{\sum_i C_i V_i}{\sum_i C_i}. \tag{6}$$

Table 5: Direct-answer test results (%). All accuracy metrics use the final reference-based ratings over the same 2,064-question panel.

Model	Accuracy	Task macro	DWG macro	Semantic
A	74.27	74.98	59.89	75.34
E	52.42	52.39	40.73	52.98
Base	37.16	33.93	42.10	38.06
InternVL2.5-8B	34.45	32.83	35.66	35.47
LLaVA-OneVision	34.54	32.28	30.47	35.00

Table 6: Common explanation-track full credit (%) under the final reference-based evaluation protocol.

Model	Answer	Evidence	Joint
Base	39.00	32.17	31.93
A	70.16	46.27	46.17
E	67.97	64.78	64.24

Table 7: E minus A under identical explanation prompting (percentage points).

Task	N	Answer difference	Joint difference
T01	440	-5.68	+38.64
T02	181	-9.39	-13.81
T03	200	-12.50	-13.50
T04	181	+6.63	+30.94
T05	181	-5.52	-4.97
T06	181	-14.36	-4.97
T07	127	-3.94	-6.30
T08	127	+42.52	+44.88
T09	200	-2.00	+46.50
T10	69	-11.59	+21.74
T11	57	+5.26	+7.02
T12	40	-7.50	+10.00
T13	40	+15.00	+80.00
T15	40	+7.50	+50.00

The denominator contains correct answers, unlike joint accuracy, which uses the complete question panel. The metric assesses observable support rather than causal faithfulness; sufficiency and comprehensiveness [25] require input interventions and new predictions and are not inferred here. Evidence validity conditional on a correct answer is 65.81% for A and 94.51% for E. Accordingly, 34.19% and 5.49% of correct answers lack full-credit evidence. This gap indicates that a correct conclusion can coexist with absent, incomplete or incorrect support; those categories should not all be called hallucination. Base attains 81.86%, but answers fewer questions correctly, so a high conditional value does not imply stronger overall supported reasoning. Conditioning on correct answers also changes the selected subset across models.

5.7 Sensitivity to Controlled Changes

Motivated by consistency analysis in GQA [15], a pair succeeds only when both answers receive full credit; merely changing an output is insufficient. The fixed inventory comprises 69 visual and 179 state pairs.

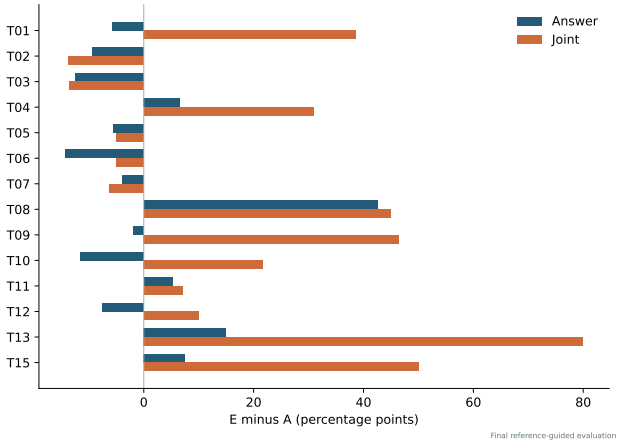


Figure 8: Task-dependent gains and losses for E. All primary families are retained.

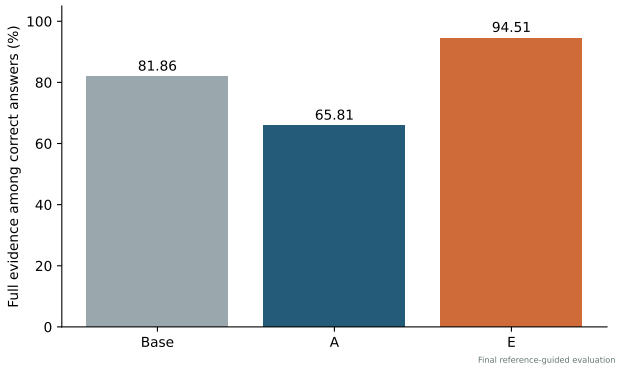


Figure 9: Evidence validity conditional on a correct answer, using the final reference-based ratings.

Shared questions and circuits preclude treating them as independent trials.

Under common explanation prompting, E succeeds on 77/82 state-flip pairs (93.90%) versus A's 55/82 (67.07%). However, E succeeds on only 11/97 state-invariant pairs (11.34%) versus A's 60/97 (61.86%). E improves on visual-invariant pairs (40/46, 86.96%, versus 36/46, 78.26%), but is weaker on visual-flip pairs (1/23, 4.35%, versus 9/23, 39.13%). Evidence-guided adaptation is therefore more successful for selected relevant state changes but insufficiently selective for irrelevant changes, and weak for topology-changing visual pairs. Better explanation production does not establish robust visual causal understanding.

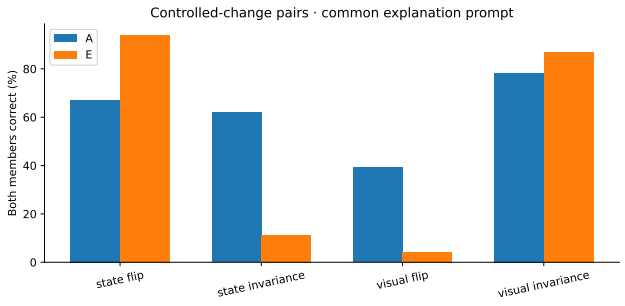


Figure 10: Both-members-correct performance on the fixed change-pair inventory.

5.8 Performance by Circuit Complexity

On the 98 questions with at least three explicit reference states, E achieves 63.27% answer accuracy versus 43.88% for A, a 19.39-point advantage. On the 1,445 two-state questions, they are close (73.08% versus 73.63%); E is worse on one-state questions (56.24% versus 69.58%). E also exceeds A on recorded paths of four to six edges (61.81% versus 58.32%) and seven or more edges (53.00% versus 49.77%).

Path lengths count edges in the local reference graph, not visible wire segments or latent reasoning steps. Questions without recorded paths are grouped separately. These exploratory comparisons do not isolate a complexity effect because task category, answer distribution, and graph complexity are correlated.

5.9 Generation Cost

Direct-track mean generation times are 0.65, 0.73 and 1.21 seconds for Base, A and E; explanation prompting increases these to 6.67, 0.82 and 2.30 seconds. Mean generated explanation-track lengths are 278.13, 13.10 and 54.81 tokens. A remains brief despite the evidence request. E's evidence advantage incurs additional generation cost. Generation times are measured per response. Efficiency statistics include all 2,074 test responses, including T14.

6 Representative Cases and Discussion

6.1 Dataset Cases as Verifiable Reasoning Examples

The appendix presents six examples from different original DWG windows, covering minimum closure, wire grouping, unknown contact states, fault diagnosis, path order, and branch interruption (Figs. 13–15). They illustrate annotation semantics rather than held-out model performance.

The examples illustrate three distinctions. Wire-only connectivity excludes state-dependent contact edges. Fault diagnosis compares observations under alternative hypotheses. Branch interruption requires tracing the queried path because a nearby contact may be irrelevant. Concise evidence makes these dependencies explicit through paths, blocking conditions, or measurement outcomes.

6.2 Interpreting Evidence Effects

Answer-only supervision produces the strongest direct-answer accuracy. Under common explanation prompting, E instead achieves higher evidence and joint accuracy, with answer advantages concentrated in measurement planning, diagnostic sufficiency and selected state-dependent tasks. These gains coexist with losses in minimum closure, grouping and bypass questions. Evidence supervision is therefore useful when supported justifications are required, but is not uniformly superior.

The original-DWG path example illustrates a successful explicit decomposition. Neither this case nor the aggregate prompt interaction proves causal faithfulness of generated text. Longer training targets, output alignment and evaluator artifacts remain alternative explanations. The fixed-seed study cannot isolate the causal contribution of evidence content.

6.3 Scope, Dependence, and Annotation Limitations

The dataset covers substation secondary wiring. Two held-out projects and 300 partly overlapping original windows limit conclusions about broader industrial transfer. Source-disjoint splits do not eliminate repeated engineering templates or isomorphic circuit graphs.

The automatic annotation pipeline can produce self-consistent but visually incorrect references if entity extraction is wrong. Source traceability facilitates correction, but cannot eliminate shared extraction errors. T12 has no verified parallel cases, T14 has a fixed negative answer prior, and generated question templates may permit shortcuts. These limitations motivate per-task reporting, label-prior controls, and explicit separation of diagnostic tasks. Native edits represent controlled perturbations rather than realistic fault prevalence or validated maintenance procedures.

6.4 Implications for Schematic Understanding

Answer-only training is preferable for overall direct-answer accuracy, whereas evidence supervision improves explicit support and selected conditional tasks. Its failures on invariant-state and topology-changing pairs indicate a need to distinguish relevant from irrelevant changes. Further evaluation should include additional drawing sources, repeated training seeds, and matched-length supervision controls.

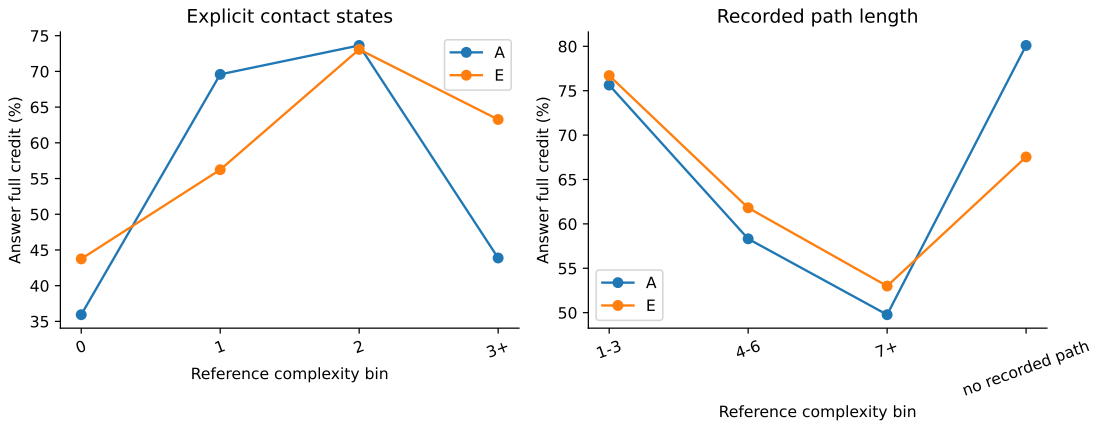


Figure 11: Answer full credit by reference-state count and maximum recorded path length. All bins are retained.

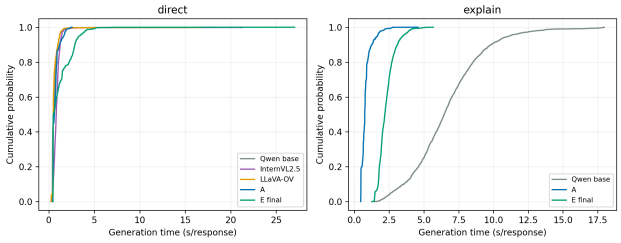


Figure 12: Generation-latency distributions under the two output protocols.

7 Conclusion

ELEC-VQA provides 10,201 electrical schematic QA pairs with concise evidence, source traceability, and graph-based verification. The comparative study shows substantial domain-adaptation gains while distinguishing answer accuracy from explanation quality. Answer-only fine-tuning achieves the strongest direct-answer accuracy. Evidence-guided fine-tuning improves joint answer–evidence accuracy and selected measurement-planning and multi-condition tasks. Failures on irrelevant-state changes and topology-changing image pairs show that explanation generation alone is insufficient for robust schematic understanding. Broader source coverage, stronger reference-quality assessment, and repeated training seeds are needed to establish generalization.

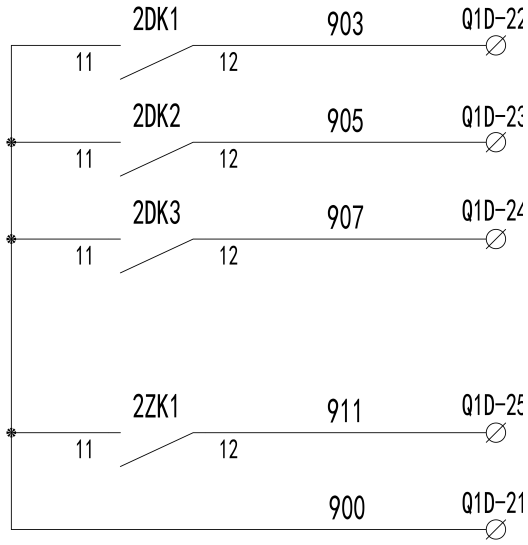
Resources

The ELEC-VQA dataset and code are available at `https://github.com/Talulah81920/ELEC-VQA`. The supplementary material provides evaluation rubrics, scoring provenance, complete task-level results, and reproducibility details: `https://github.com/Talulah81920/ELEC-VQA/blob/main/paper/Supplementary_Material.pdf`.

Appendix A Dataset Examples

A.1 Examples from Original Schematic Windows

Figures 13–15 show six training examples drawn from six different original DWG windows, with no controlled wiring edits. None reuses a screenshot displayed in the main text. Questions and short answers retain the released wording; concise evidence summaries highlight the corresponding graph checks. These examples document dataset content, not model predictions or observed field faults. The examples retain their original source associations.

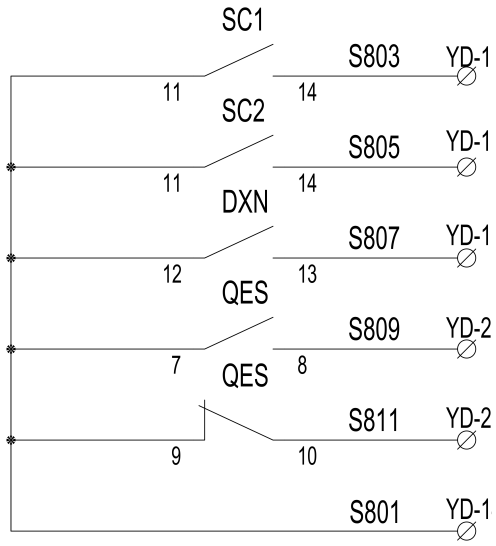


(a) Minimum contact closure

Question
Which contacts must close to get continuity from 2DK1:12 to 2ZK1:12? Give every combination that works without extra contacts; leave the others open. Choose from 2DK1 contact 11–12, 2DK2 contact 11–12, 2DK3 contact 11–12 and 2ZK1 contact 11–12. Treat unlisted contacts as open.

Answer
2DK1 contact 11–12 and 2ZK1 contact 11–12 must be closed.

Evidence summary
2DK1:12 → 2DK1:11 → 2ZK1:11 → 2ZK1:12 requires both 2DK1 11–12 and 2ZK1 11–12. Opening either selected contact breaks continuity; no other minimal closure set works.



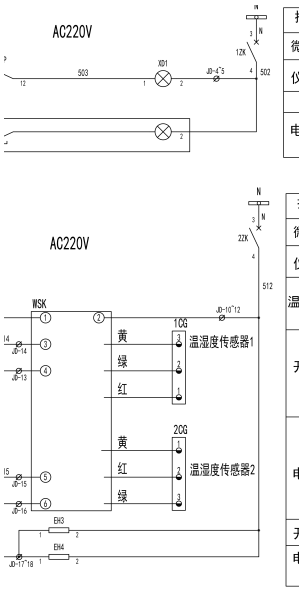
(b) Wire-only terminal groups

Question
Which of these terminals share a wire-only connection: DXN:12, DXN:13, QES:10, QES:7, QES:8, QES:9, SC1:11, SC1:14? Keep contact gaps separate and include isolated terminals.

Answer
{DXN:12, QES:7, QES:9, SC1:11}; {DXN:13}; {QES:10}; {QES:8}; {SC1:14}.

Evidence summary
External wires join DXN:12, QES:7, QES:9 and SC1:11. The other listed terminals form separate wire-only groups; contact gaps do not merge these groups.

Figure 13: Minimum contact closure and wire-only terminal grouping. Contact conduction is governed by the stated assumptions, whereas wire groups exclude contact edges.

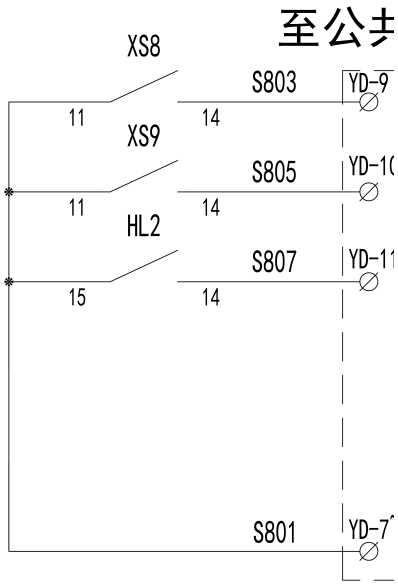


(c) Unknown contact state

Question
Is the connection between 2ZK:3 and EH4:2 fixed, or does it depend on 2ZK contact 3–4? Assume 1ZK contact 3–4 closed. Treat unlisted contacts as open.

Answer
It depends on that contact: connected when closed, disconnected when open.

Evidence summary
Closing 2ZK 3–4 gives 2ZK:3 → 2ZK:4 → EH4:2. Opening that contact disconnects the pair under the stated assumptions.



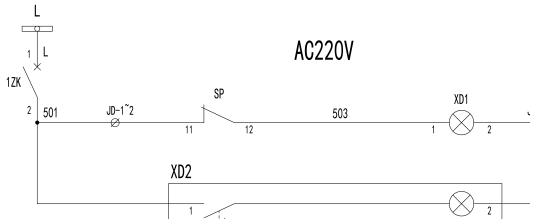
(d) Fault-hypothesis discrimination

Question
In an isolated continuity check from HL2:14, the readings are HL2:15 connected and XS8:14 connected. Exactly one of XS9 contact 14–11, XS8 contact 14–11 and HL2 contact 14–15 is stuck open; all other named contacts are closed and wires are intact. Which of these contacts could be stuck open? Treat unlisted contacts as open.

Answer
XS9 contact 14–11.

Evidence summary
Predicted readings (HL2:15, XS8:14): XS9 fault → (1,1); XS8 fault → (1,0); HL2 fault → (0,0). Only the XS9 fault matches both observations.

Figure 14: State dependence and fault-hypothesis discrimination. Different contact states and fault hypotheses are checked against their corresponding continuity outcomes.

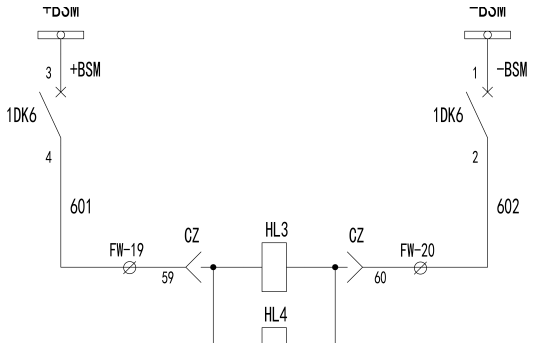


(e) Contact order along a route

Question
From 1ZK:1 to XD1:1, in what order does a conducting path cross the contacts? Assume 1ZK contact 1–2 and SP contact 11–12 closed. Treat unlisted contacts as open.

Answer
1ZK contact 1–2 → SP contact 11–12.

Evidence summary
1ZK:1 → 1ZK:2 → SP:11 → SP:12 → XD1:1. The route traverses 1ZK 1–2 before SP 11–12.



(f) Branch-selective interruption

Question
Will opening 1DK6 contact 1–2 interrupt the path from 1DK6:3 to CZ:59? Initially, 1DK6 contact 3–4 and 1DK6 contact 1–2 closed. Other contacts keep their stated states. Treat unlisted contacts as open.

Answer
No.

Evidence summary
The route 1DK6:3 → 1DK6:4 → CZ:59 remains available after 1DK6 1–2 opens. That contact is outside the queried conducting path.

Figure 15: Contact order and branch-selective interruption. A valid route identifies the traversed contacts; an unrelated contact opening leaves the queried branch connected.

A.2 Annotation and Evaluation Records

Each annotation contains a question, short answer, compact evidence, typed result, task parameters, and source provenance. Verification queries store contact states, graph modifications, connectivity outcomes, and supporting paths or reachable regions. The two training views serialize the same image–question records with different supervision targets; they do not increase the number of distinct QA examples.

Answer and evidence quality are evaluated separately with a three-level electrical rubric. All responses are retained in the final scoring set, and joint correctness requires both dimensions to receive full credit for the same response.

A.3 Structural Coverage and Pair Inventory

Figure 16 summarizes the bounded reference graphs for the 300 original context windows, excluding edited variants. These are graph-level counts, not an inventory of every component visible in each screenshot. Table 8 records the completed binary pair audit. The inventory is restricted to binary decisions with revalidated outcomes.

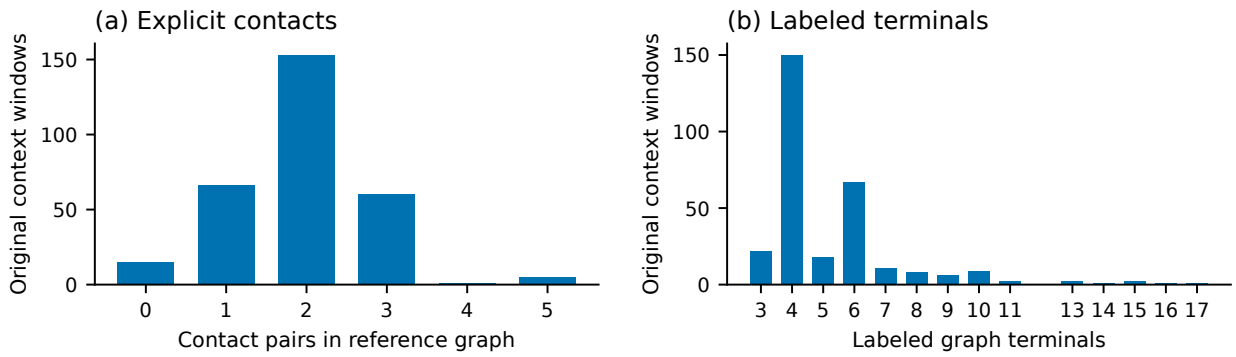


Figure 16: Structural coverage of the bounded reference graphs in original contexts. Original windows may overlap within a drawing; graph statistics do not imply independent circuit designs.

Table 8: Revalidated binary pair inventory. Counts reuse existing QA and are not model scores.

Split	Visual flip	Visual invariant	State flip	State invariant
Train	79	178	340	283
Validation	13	23	37	58
Test	23	46	82	97

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